ELECTRICAL POWER GENERAL NOTES

1. REMOVE ALL UNUSED CONDUIT, FLEXIBLE CABLE AND COMMUNICATION CABLES FROM ABOVE CEILING.
2. PROVIDE NEW TYPED INDEX CARD FOR EACH PANELBOARD THAT IS MODIFIED. INDEX CARD SHALL REFLECT LOAD SERVED, ROOM NUMBER AND OR WORKSTATION NUMBER.
3. CONTRACTOR SHALL PROVIDE AN AS-BUILT SET OF BLUEPRINTS AT JOB COMPLETION.
4. CONTRACTOR SHALL NOT SHARE PC NEUTRALS WITH ANY GENERAL PURPOSE CIRCUITS.
5. ALL CIRCUIT NUMBERS SHOWN ARE FOR REFERENCE ONLY. FIELD VERIFY EXACT CIRCUITS USED.
6. CONTRACTOR SHALL MARK ALL CIRCUITS THAT ARE NOT USED DUE TO REMODELING IN EXISTING PANEL BOARD DIRECTORY "SPARE".
7. PROVIDE CONTINUNITY TO EXISTING OUTLETS THAT REMAIN WHERE POWER WAS DISCONNECTED DUE TO THE REMODELING.
8. ALL CABLE RUNS ABOVE CEILING SHALL BE PLENUM RATED.
9. INSTALL ADJACENT OUTLETS AT 6" ON CENTER.
10. ALL RECEPTACLES MOUNTED WITH-IN 6'-0" OF SINK SHALL BE "GFCI" TYPE.
11. ALL WORK IN THIS LEASE SHALL COMPLY WITH BASE BUILDING SPECIFICATIONS.
12. REFER TO ARCHITECTURAL DRAWINGS FOR EXACT LOCATIONS AND MOUNTING HEIGHTS OF ALL WIRING DEVICES.
13. ALL WIRING DEVICES ON A DEDICATED CIRCUIT SHALL BE A NEMA #5-20R. CONTRACTOR SHALL REMOVE ALL EXISTING FLOOR OUTLETS THAT ARE NOT BEING RE-USED.
14. PATCH EXISTING CORE HOLES WITH CONCRETE TO MAINTAIN FIRE RATING. ELECTRICAL CONTRACTOR SHALL VERIFY ALL EXISTING FIELD CONDITIONS PRIOR TO SUBMITTING A BID.
15. ELECTRICAL CONTRACTOR SHALL REMOVE ALL UNUSED WIRING, CONDUIT, WIRE, ETC., NOT SHOWN ON DRAWINGS TO REMAIN. REMOVE UNUSED CIRCUITS, CONDUIT AND WIRE BACK TO EXISTING PANELS OR NEAREST HOT OUTLET WHICH REMAINS TO PROVIDE CONTINUITY OF EXISTING CIRCUITS.
16. ALL FLEXIBLE/RIGID CONDUIT AND COMMUNICATION/DATA CABLING ROUTED ABOVE THE CEILING SHALL BE INDEPENDENTLY SUPPORTED FROM HANGER WIRES. THE LAYING OF FLEXIBLE/RIGID CONDUIT AND COMMUNICATION/DATA CABLING ON CEILING TILES OR USING SUSPENDED CEILING HANGER WIRES FOR SUPPORTING MEANS IS UNACCEPTABLE
17. AFTER DEMOLITION IS COMPLETE CONTRACTOR SHALL TRACE ALL EXISTING CIRCUITS REMAINING IN ALL PANEL BOARDS TO CONFIRM LOAD SERVED. IF EXISTING CIRCUITS DO NOT SERVE ANY LOAD ALL WIRING SHALL BE COMPLETELY REMOVED BACK TO EXISTING CIRCUIT BREAKER. IF EXISTING CIRCUIT SERVES A LOAD THIS LOAD SHALL BE RECORDED ON THE NEW TYPED INDEX CARD.
18. ALL DUPLEX AND QUADRAPLEX RECEPTACLES TO BE WIRED WITH 3#12 COPPER WIRE ROUTED IN 3/4" CONDUIT ABOVE CEILING TO DESIGNATED PANEL BOARD AND CIRCUIT BREAKER UNLESS NOTED OTHERWISE.
19. ALL DEDICATED CIRCUITS SHALL BE SEPARATELY CIRCUITED WITH 3#12, IN A 3/4" CONDUIT BACK TO DESIGNATED PANEL BOARD AND CIRCUIT BREAKER UNLESS NOTED OTHERWISE.

ELECTRICAL LIGHTING GENERAL NOTES

1. REFER TO ARCHITECTURAL DRAWINGS FOR EXACT LOCATION AND QUANTITY OF ALL LIGHT FIXTURES.
2. ALL CIRCUIT NUMBERS SHOWN ARE FOR REFERENCE ONLY. FIELD VERIFY EXACT CIRCUITS USED AND REVISE PANEL SCHEDULE ACCORDINGLY.
3. ALL SWITCHES GANGED TOGETHER SHALL BE MOUNTED UNDER A COMMON COVERPLATE AND SHALL MATCH EXISTING BUILDING COLOR.
4. CLEAN ALL EXISTING AND RELOCATED LIGHT FIXTURES AND REPLACE ALL LAMPS WITHIN AREA OF WORK.
5. ALL LIGHT SWITCHES SHALL BE "DECORATOR" SERIES WHITE ON WHITE ROCKER STYLE SWITCHES U.N.O.
6. REMOVE ALL EXISTING SWITCHES THAT DO NOT MATCH NEW AND REPLACE WITH NEW WHITE ON WHITE SWITCHES "DECORATOR" SERIES.
7. WIRE AND CONNECT ALL 277 VOLT LIGHTING CIRCUITING TO EXISTING ABOVE CEILING LIGHTING GRID.
8. THE CONTRACTOR SHALL UPDATE ALL PANELBOARD DIRECTORIES AFTER CONSTRUCTION IS COMPLETE. CONTRACTOR SHALL FURNISH MARKED-UP AS-BUILT PRINTS SHOWING EXACT CIRCUIT NUMBERS USED AND ALL LIGHT FIXTURES THAT HAD TO BE REMOVED OR RELOCATED DUE TO JOB SITE CONFLICTS.
9. ALL FLEXIBLE/RIGID CONDUIT AND COMMUNICATION/DATA CABLING ROUTED ABOVE THE CEILING SHALL BE INDEPENDENTLY SUPPORTED FROM HANGER WIRES. THE LAYING OF FLEXIBLE/RIGID CONDUIT AND COMMUNICATION/DATA CABLING ON CEILING TILES OR USING SUSPENDED CEILING HANGER WIRES FOR SUPPORTING MEANS IS UNACCEPTABLE.

KEYED NOTES

1 EXISTING WIRING DEVICE TO REMAIN.

2 PROVIDE 3/4" CONDUIT UNDER RAISED FLOOR FOR POWER WIRING TO FLUSH FLOOR DEVICES. PROVIDE 1" CONDUIT RAISED UNDER FLOOR FOR DATA/COMMUNICATION CABLING TO FLUSH FLOOR DEVICES.

3 3/4" CONDUIT RISE FROM RAISED FLOOR BELOW AND POWER WIRING HOMERUN TO PANEL. 1" CONDUIT RISE FROM BELOW RAISED FLOOR AND ELBOW IN CEILING SPACE FOR DATA CABLING. PROVIDE RUBBER GROMMIT AND PULL STRING FROM FLUSH FLOOR DEVICES.

4 EXISTING LIGHTING AND SWITCHING TO REMAIN IN THIS AREA.

5 THIS SWITCH SHALL OVERRIDE CEILING MOUNTED OCCUPANCY SENSOR IN ORDER TO MANUALLY TURN LIGHTS OFF. WHEN THE SWITCH IS IN THE ON POSITION, THE CEILING MOUNTED OCCUPANCY SENSOR WILL TURN THE LIGHTS ON AND OFF AUTOMATICALLY.

6 CONNECT TO EXISTING 277V LIGHTING CIRCUIT IN THE AREA.

7 LIGHTING FIXTURES TO BE WIRED TOGETHER IN THIS ROOM WITH FLEXIBLE MC CABLE AND CONTROLLED BY SWITCH OR OCCUPANCY SENSOR LOCATED IN THIS ROOM. EACH FIXTURE TO BE WIRED WITH 1#12 COPPER WIRING AND 1#12 NEUTRAL IN MC CABLE. FLEXIBLE MC CABLE SHALL BE LIMITED TO 20'-0 LENGTHS. WIRING IN RUNS LONGER THAN 20'-0 SHALL BE INSTALLED IN RIGID EMT CONDUIT. FURNISH AND INSTALL NEUTRAL BETWEEN OCCUPANCY SENSOR AND LIGHT FIXTURES.

8 JUNCTION BOX TO SERVE UNDERCABINET TASK LIGHTING.

9 CONNECT TO EXISTING 120V RECEPTACLE CIRCUIT IN THE AREA.

10 PROVIDE NEW TRACK LIGHTING HEAD AND CONNECT TO EXISTING TRACK FIXTURE. RE; LIGHTING FIXTURE SCHEDULE FOR MORE INFORMATION. ADJUST ALL TRACK HEADS AND FOCUS AS DIRECTED BY TENANT.

11 JUNCTION BOX FOR LSI VISTA VU DISPLAY.

12 JUNCTION BOX FOR BACKLIT LOGO SIGN.

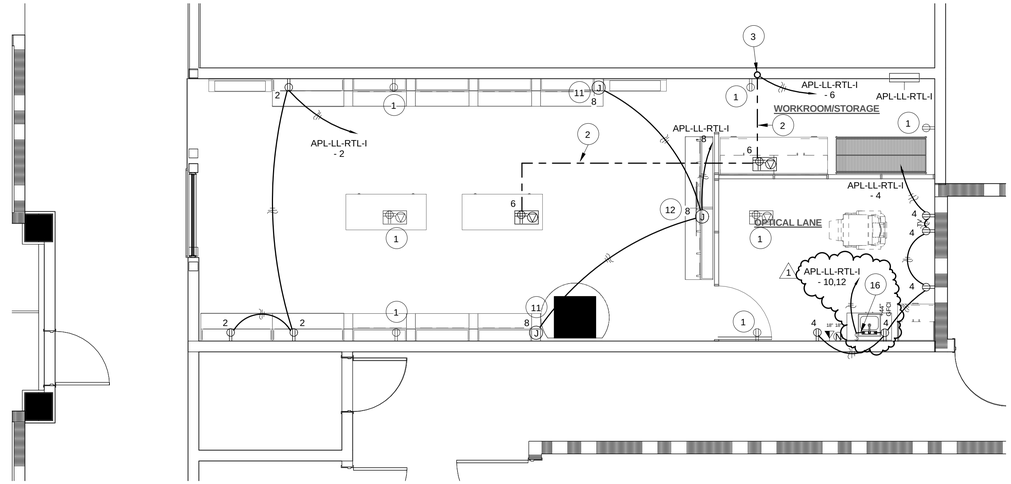
13 EXISTING SPEAKER TO REMAIN.

14 ~~DISCONNECT AND REMOVE EXISTING SPEAKER.~~

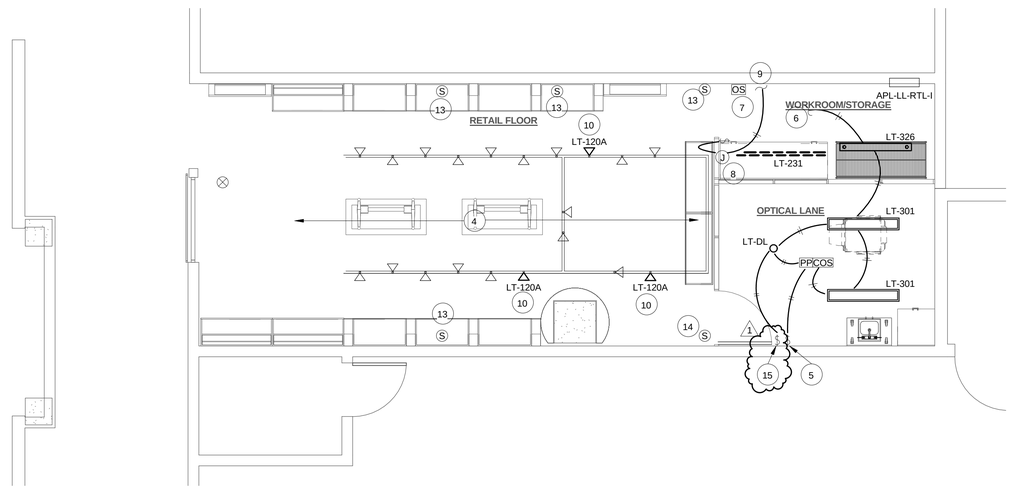
15 LUTRON NOVA T-STAR LINEAR SLINE ON/OFF FLUORESCENT LIGHT SWITCH #NTF-10-277 WHITE AND MOUNTED BEHIND A COMMON COVERPLATE WITH OTHER SWITCHES. DIMMER SWITCH SHALL SERVE AS AN OVERRIDE SWITCH TO THE CEILING MOUNTED OCCUPANCY SENSOR. 1

16 NEW 30A-2P NF (LEVITON #MS302-DS OR EQUAL) TOGGLE TYPE DISCONNECT SWITCH IN LOCKABLE ENCLOSURE #N13NC-TDS MOUNTED BELOW COUNTER TOP FOR CONNECTION TO POINT OF USE ELECTRIC WATER HEATER.

REVISION
10/03/14
S. TOVAR



1 ELECTRICAL POWER PLAN LOWER LEVEL
1/4" = 1'-0"



2 ELECTRICAL LIGHTING PLAN LOWER LEVEL
1/4" = 1'-0"

ISSUES

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1	09/24/14	REVISION 1

STATE OF TEXAS
MATTHEW J. MEADOR
113101
LICENSED PROFESSIONAL ENGINEER
09/24/14

Burns, DeLotte & McCoy, Inc.
Texas Registered Engineering Firm
F-685

Browne McGregor

Browne McGregor
Architects, Inc.

520 Post Oak Boulevard
Suite 880
Houston, Texas 77027
713 850-1733 Phone
713 850-0833 Fax
www.brownemcgregor.com

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HOUSTON EYE ASSOCIATES

EXXONMOBIL CAMPUS - NATURE 2 LOWER LEVEL
MERCANTILE RETAIL SPACE
SPRING, TX

OWNER:
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Job No.:	001612.00
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Sheet Name
ELECTRICAL FLOOR PLANS LOWER LEVEL

Sheet No.:
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EXISTING

PANEL LOAD SCHEDULE

LOCATION: WORKROOM/STORAG...
PROVIDE ENGRAVED NAME PLATE

PANEL: **APL-LL-RTL-I** VOLTAGE: 120/208 VOLTS - 3 PHASE - 4 WIRE MAINS: 100 A MCB: 100A-3P TYPE: Surface

LOAD	WIRE	TRIP	P	CCT	A	B	C	A	B	C	CCT	P	TRIP	WIRE	LOAD
EXISTING LOAD	--	20 A	1	1	0 VA			540 VA			2	1	20 A	#12	3 REC
EXISTING LOAD	--	20 A	1	3		0 VA			970 VA		4	1	20 A	#12	5 REC
EXISTING LOAD	--	20 A	1	5			0 VA			1000 VA	6	1	20 A	#12	4 REC - FLOOR
EXISTING LOAD	--	20 A	1	7	0 VA			600 VA			8	1	20 A	#12	DISPLAY CASES
EXISTING LOAD	--	20 A	1	9		0 VA			6000 VA		10	2	40 A	#8	EWH-1 1
EXISTING LOAD	--	20 A	1	11			0 VA			0 VA	12				
EXISTING LOAD	--	20 A	1	13	0 VA			0 VA			14	1	20 A	--	SPARE
SPARE	--	20 A	1	15		0 VA			0 VA		16	1	20 A	--	SPARE
SPARE	--	20 A	1	17			0 VA			0 VA	18	1	20 A	--	SPARE
SPARE	--	20 A	1	19	0 VA			0 VA			20	1	20 A	--	SPARE
SPARE	--	20 A	1	21		0 VA			0 VA		22	1	20 A	--	SPARE
SPARE	--	20 A	1	23			0 VA			0 VA	24	1	20 A	--	SPARE
SPARE	--	20 A	1	25	0 VA			0 VA			26	1	20 A	--	SPARE
SPARE	--	20 A	1	27		0 VA			0 VA		28	1	20 A	--	SPARE
SPARE	--	20 A	1	29			0 VA			0 VA	30	1	20 A	--	SPARE
SPARE	--	20 A	1	31	0 VA			0 VA			32	1	20 A	--	SPARE
SPARE	--	20 A	1	33		0 VA			0 VA		34	1	20 A	--	SPARE
SPARE	--	20 A	1	35			0 VA			0 VA	36	1	20 A	--	SPARE
SPARE	--	20 A	1	37	0 VA			0 VA			38	1	20 A	--	SPARE
SPARE	--	20 A	1	39		0 VA			0 VA		40	1	0 A	--	SPARE
SPARE	--	20 A	1	41			0 VA			0 VA	42	1	20 A	--	SPARE

ELECTRICAL LOAD ANALYSIS:

	A	B	C
TOTAL VOLT-AMPERES	1140 VA	6970 VA	1000 VA
AMPERES PER PHASE	10 A	58 A	8 A
1. TOTAL NEW LOAD:	25 A		
2. +25% NEW LOAD TOTAL:	6 A		
3. EXISTING LOAD:	4 A		
4. TOTAL LOAD:	35 A		

NOTES: EXISTING PANEL FED FROM 100A-3P BREAKER IN DISTRIBUTION PANEL DPL-LL-RTL.

SPECIFICATIONS

THIS PROJECT TO BE PERFORMED IN ACCORDANCE WITH BASE BUILDING SPECIFICATIONS. BIDDING CONTRACTOR TO RECEIVE COPY OF SPECIFICATIONS AT TIME OF BIDDING FROM EXXONMOBIL. WHERE CONFLICT EXISTS BETWEEN THE DRAWINGS AND THE BASE BUILDING SPECIFICATIONS, THE MOST STRINGENT SHALL GOVERN. MAKE THE ENGINEER AND OWNER AWARE IN WRITING WHENEVER A CONFLICT EXISTS BETWEEN THE DRAWINGS AND BASE BUILDING SPECIFICATIONS.

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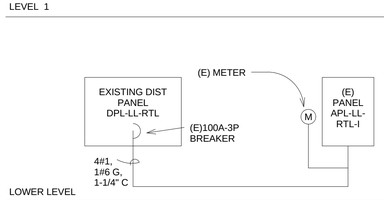
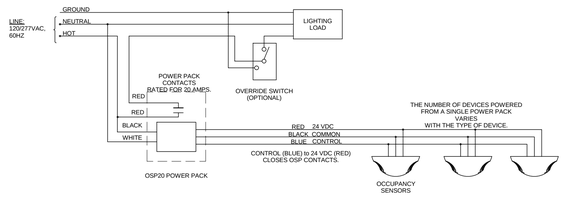


1 One-Line Diagram
N.T.S.



2 Detail - Ceiling Occupancy Sensor Detail
N.T.S.

ELECTRICAL LEGEND

SYMBOL	DESCRIPTION
$	NEW SINGLE POLE "DECORATOR" SERIES, ROCKER TYPE LIGHT SWITCH, WHITE-ON-WHITE, MOUNTED +48" AFF U.N.O. TO CENTER OF BOX.
	NEW QUADRAPLEX-TWO-NEMA #5-2OR WHITE DUPLEX RECEPTACLES MOUNTED BEHIND A COMMON COVERPLATE. MOUNT WIRING DEVICE AT 18" AFF. FROM CENTER OF BOX. U.N.O. ENGRAVE BOTTOM OF COVER PLATE WITH PANEL NAME AND CIRCUIT NUMBER.
	NEW NEMA #5-20R DUPLEX RECEPTACLE, "DECORATOR" STYLE, WHITE ON WHITE. MOUNTED +18" AFF. FROM CENTER OF BOX U.N.O. ON ARCHITECTURAL DRAWINGS. ENGRAVE BOTTOM OF COVERPLATE WITH PANEL NAME AND CIRCUIT NUMBER.
	NEW COMBINATION COMMUNICATION/DATA OUTLET MOUNTED +18" AFF. FROM CENTER OF BOX U.N.O. PROVIDE BACK BOX AND PULL STRING TO ABOVE CEILING. CABLING CONTRACTOR SHALL INSTALL ALL DATA CABLING AND INSTALL WHITE COVERPLATE.
OS	NEW LEVITON #OSSMT-MDW MULTI-TECHNOLOGY WALL MOUNTED OCCPANCY SENSOR WITH PASSIVE INFARED/ULTRASONIC TECHNOLOGY, WHITE ON WHITE WITH MANUAL ON/AUTO OFF AND 15 SECOND-30 MINUTE TIME DELAY. CONTRACTOR SHALL ADJUST OCCUPANCY SENSOR TO MAXIMUM SETTING.
COS	NEW LEVITON #OSC20-MOW MULTI-TECHNOLOGY CEILING MOUNTED OCCUPANCY SENSOR WITH PASSIVE INFARED/ULTRASONIC TECHNOLOGY, WHITE ON WHITE WITH SENSITIVITY ADJUSTMENT AND 15 SECOND-30 MINUTE TIME DELAY. CONTRACTOR SHALL ADJUST OCCUPANCY SENSOR TO MAXIMUM SETTING.
PP	DUAL RELAY TYPE SENSOR SWITCH PACK MOUNTED ABOVE CEILING ON JUNCTION BOX, LEVITON #OSP20. WIRE AND CONNECT TO CIRCUIT AS REQUIRED.
	NEW WALKER RC4AGY-AC-C5EKEY FULLY ASSEMBLED, FLUSH MOUNTED FIRE RATED POKE THRU WITH TWO NEMA #5-20R DUPLEX RECEPTACLES 4 RECESSED MOUNTED CATEGORY 6E KEYSTONE, JACKS FOR COMMUNICATION CAPABILITIES. COORDINATE EXACT LOCATION WITH BUILDING STRUCTURAL MEMBERS PRIOR TO CORING. X-RAY FLOOR SLAB AS REQUIRED.
OR J	JUNCTION BOX
	LIGHTING & RECEPTACLE PANELBOARD.
TV	NEW LEVITON NO. TV1WW RECESSED DUPLEX RECEPTACLE WITH A/V FOR TV. PROVIDE CONDUIT AND WIRING TO ABOVE CEILING SPACE AND ELBOW FOR DATA BOX WITH PULL STRING. A/V CONTRACTOR TO PROVIDE A/V COVER PLATE AND A/V WIRING. COORDINATE MOUNTING HEIGHT WITH ARCHITECTURAL PLANS.
	EXISTING EXIT LIGHT FIXTURE TO REMAIN.
AFF.	ABOVE FINISHED FLOOR
(R)	EXISTING ITEM RELOCATED
(E)	EXISTING TO REMAIN.
(N)	NEW.
U.N.O.	UNLESS NOTED OTHERWISE.

LIGHTING FIXTURE SCHEDULE

SYMBOL	DESCRIPTION
LT-326	COLUMBIA KL-4-2-32-VR-U SUSPENDED FIXTURE. 277V F32T8/TL830/PLUS/ALTO (QTY 2) LAMPS.
LT-301	MARK LIGHTING SL4F-TG-1-T5H0-EDPR-277-FA RECESSED STRIP FLUORESCENT LIGHT FIXTURE. 277V.
1 LT-DL	LITHONIA 6HF-1/26-42TRT-F6B5W-277 FLUORESCENT DOWNLIGHT FIXTURE. 277V CFL. FURNISH THIS FIXTURE WITH LUTRON HI-LUME, 277V, 100% TO 0% DIMMING BALLAST.
LT-231	PHILLIPS COLOR KINETICS EW PROFILE POWERCORE LED UNDERCABINET LIGHT FIXTURE. LED WHITE 3000K 120V. WITH INTEGRAL ROCKER SWITCH.
LT-120A	LIGHTING SERVICES INC. VLX2044F3M400 LED TRACK HEAD MOUNTED ON LT-337. 25W LED 3000K. MOUNT ON EXISTING TRACK FIXTURE.

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1	09/24/14	REVISION 1

STATE OF TEXAS
MATTHEW J. MEADOR
113101
LICENSED PROFESSIONAL ENGINEER
09/24/14

Burns, DeLotte & McCoy, Inc.
Texas Registered Engineering Firm
F-685

Browne McGregor

Browne McGregor
Architects, Inc.

520 Post Oak Boulevard
Suite 880
Houston, Texas 77027
713 850-1733 Phone
713 850-0833 Fax
www.brownemcgregor.com

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MERCANTILE RETAIL SPACE
SPRING, TX

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ELECTRICAL NOTES AND SCHEDULES

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